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Evaluating the relationship between income inequality, renewable energy and energy poverty in the V4 countries

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ABSTRACT

The European Union has prioritized addressing energy poverty, as outlined in the "Clean Energy for all Europeans Package". The recent international context dominated by the COVID-19 pandemic and Russia war in Ukraine, have exacerbated this issue because of the substantial energy prices growth. Since eradicating energy poverty necessitates long-term policy implementation, the primary objective of this study is to evaluate the impact of income inequality and other economic and social factors (output per capita, renewable energy use) on energy poverty as measured by three indicators (total utility bill arrears, utility bill arrears for households with dependent children, population unable to maintain adequate). The findings based on mean group estimators for the V4 (Poland, Hungary, Czech Republic, Slovakia) countries from 2005 to 2022 suggest that income inequality, proxied by the Gini index, exacerbates energy poverty. However, sustainable economic growth has the potential to alleviate energy poverty. Policy recommendations include reducing income inequality, reforming renewable energy policies, to encourage the development of renewable energy sector and green industries.

1. Introduction

Lately, energy poverty has emerged as a pressing issue in the European Union, affecting over 50 million households [1]. This phenomenon, characterized by the inability to afford adequate energy services, poses significant social and economic challenges. The recent epidemic and the war in Ukraine have exacerbated energy poverty, highlighting the need for comprehensive long-term solutions. The countries near the conflict area were more exposed to this issue.

Ending poverty is one of the global goals [2,3]. This was also decreed with the publication of the 2030 Agenda and the Sustainable Development Goals in 2015 [4–6]. The United Nations approved 17 Goals, including eradicating poverty (SDG1), and achieving affordable and clean energy (SDG7). Policymakers must deliver policies that contribute to a more sustainable world [7–9].

This study delves into the intricate income inequality-energy poverty nexus within the V4 (Poland, Hungary, Czech Republic, Slovakia) countries context, spanning the period from 2005 to 2022. While previous research has established a correlation between these two factors (for example, Igawa and Managi showed negative correlation in 37 states [10], Nguyen and Nasir showed negative correlation in 51 states during 2001–2014 [11], this study goes beyond mere association to investigate the mechanisms through which income inequality contributes to energy poverty. High income inequality concentrates wealth in a smaller group. This leaves low-income households with less disposable income after covering basic needs like food and rent. When faced with limited resources, energy becomes a non-essential expense that gets sacrificed. Households might reduce energy consumption and accumulate utility bill arrears. Households characterized by low-income often have old and weak energy-efficient buildings. They may struggle to afford energy-saving upgrades, further increasing their energy costs. Moreover, there are no existing studies on V4 countries, offering with this paper new findings to the literature on income disparity-energy poverty nexus. The previous studies refer to entire EU, but not to the separate analysis of the group of V4 countries [12,13].

A panel data analysis utilizing specific econometric techniques reveals that income inequality indeed had a significant effect on energy poverty. This finding aligns with the notion that low-income households, typically spending a disproportionately large share of the money spent on energy, are more susceptible to this phenomenon.

Intriguingly, the study also uncovers a potential paradoxical renewable energy use-energy poverty nexus. While renewable energy is

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widely recognized as a sustainable and environmentally friendly energy source [14,15], its rapid deployment can inadvertently contribute to energy poverty by driving up energy prices. This observation underscores the need for careful policy formulation to ensure that renewable energy transition does not exacerbate energy poverty.

To address this complex challenge, the study proposes a set of policy recommendations aimed at mitigating energy poverty while promoting sustainable energy development. These recommendations include: reducing income inequality, implementing policies that enhance human capital through education and training, promoting equitable access to employment opportunities, and addressing gender pay disparities can effectively reduce income inequality and consequently alleviate energy poverty. Designing renewable energy policies that minimize the burden of renewable taxes and surcharges on end-users can shield vulnerable households from the potential price spikes associated with renewable energy transition [16,17]. Renewable energy transitions can initially be more expensive for consumers. However, they could acknowledge the need for a fairer distribution of costs. Households with low-income are not influenced in the same manner by energy price growth. A tiered pricing structure could be explored. High-energy users, like some industries or frequent flyers, could pay a larger share of the renewable energy costs compared to low-income households. Air travel is an important driver of pollution. Implementing a surcharge on airplane tickets specifically to fund renewable energy could be a way to make this sector contribute more towards a sustainable future. A small surcharge on airline tickets could generate significant revenue to support the transition to renewables, without heavily impacting individual passengers. Using EU funds in renewable energy projects could enforce the deployment of sustainable energy sources while generating economic benefits that can contribute to alleviating energy poverty.

The study recognizes that addressing energy poverty requires a multifaceted approach that encompasses not only income inequality but also broader social, economic, and environmental factors. Future research should further explore the bidirectional relationship between income inequality and energy poverty, examining how energy poverty can exacerbate income inequality and inversely. Additionally, studies should delve into the effectiveness of targeted energy poverty interventions tailored to specific demographic groups and socioeconomic contexts.

In conclusion, the novelty of this paper relies on the connection between energy and social issues in the area of V4 countries that was not subject to other studies. This paper comes up with relevant insights into the intricate relationship between income inequality, renewable energy, and energy poverty in the V4 countries. By highlighting the complex interplay of these factors, the study underscores the necessity for reliable and multifaceted policies that consider the root causes of energy poverty and the potential unintended consequences of renewable energy transition. Energy poverty represents a complex concept, but there are three main root causes: poor income, huge energy prices and weak energy efficiency [13]. First, people with low incomes may struggle to afford even vital needs in terms of energy. This can force them to cut back on energy use, which can have negative effects on their health status and their well-being. Second, even with a moderate income, people may suffer of energy poverty if the cost of energy is high. This can be due to factors like volatile global energy markets or local monopolies. Third, poor insulated homes or these with inefficient appliances use more energy to maintain comfortable temperatures. This can lead to higher energy bills for residents, even if they are trying to conserve energy. Compared to other types of studies [13,17], this paper designs specific policies that simultaneously tackle energy crisis and social issues. Only through a holistic approach can the V4 region effectively tackles this pervasive issue and ensure that its energy transition advantages all its citizens.

The paper is based on more sections. Consequently, Section 2 presents a literature review on energy poverty; Section 3 shows Data and methodology; then Section 4 the results and discussion, and finally, Section 5 shows the main conclusions.

2. Literature review

This section explores the concept of "energy poverty", their measures, and the factors that influence it. Based on empirical evidence, hypotheses are developed to explain the impact of various factors on energy poverty.

Energy poverty is a pressing issue in the European Union, with over 100 million people unable to afford adequate indoor warmth. The ongoing conflict in Ukraine is exacerbating energy poverty, creating a more challenging environment. The V4 countries that are close to Ukraine and presents more vulnerabilities on energy poverty.

The definition of this concept remains contested, as there is no universally accepted consensus on its meaning. Historical perspectives have linked energy poverty to "fuel poverty," which represents the inability to maintain enough warmth in the home [18]. However, some researchers consider the terms synonymous, while others distinguish them [19].

Early studies linked energy poverty to household income, highlighting the vulnerability of low-income households to afford energy expenditures [20]. This approach was prevalent in developed countries that proposed expenditure-based indicators to compute the energy poverty.

Most studies define energy poverty as no access to modern energy facilities, like clean cooking, electricity, and commercial energy [21]. This definition is further expanded by Parajuli, who connects it to broader social and economic development goals [22].

From this perspective, energy poverty is seen as a concept with more dimensions: a subjective component based on household members' perception of inadequate warmth during cold periods and an objective component based on the net income of a household after energy expenditures falling under the state poverty line and proper energy costs exceeding the average level for that country [23].

Multidimensional factors that influence energy poverty include more indicators on social, economic, and environmental development. Boardman identifies economic and social factors that characterize the household, energy prices, and efficiency as key multidimensional factors [21]. Recalde et al. emphasize the importance of structural determinants of energy poverty to encompass the multidimensional perspective: economic, labor, and market policies, and energy housing [24].

Nussbaumer et al. proposed an energy poverty index based on the IEA Energy Development Index [25]. This index measures a household's inability to have energy but also considers economic and social aspects beyond accessibility, affordability, and use of energy for various necessities [26].

Energy poverty is computed by employing three approaches: physical, technological, and economic threshold. The physical threshold approach aligns with the World Bank's definition of absolute poverty, establishing a minimum level of energy use to ensure primary needs [22].

The technological threshold version focuses on the impossibility of reaching energy services, while the economic method establishes the highest percentage of income necessary for an acceptable quantity of energy expenditure. These approaches face limitations, such as the lack of available datasets to measure energy consumption for basic necessities, differences in measuring adequate temperature, and disparities between energy prices and purchasing power.

Empirical studies in literature propose various indicators to measure energy poverty depending on data availability. Some studies focus on access to different types of energy, while others focus on access to electricity, modern cooking technologies, modern energy fuels, adequate heating, arrears on energy bills, health issues generated by inadequate heating possibilities, dwelling with leakages and damp walls, and unavailability of other basic necessities. For example, Pachauri et al. proposed a composite index to compute energy poverty that combines the type of access to energy and the volume of energy

[27]. Distribution of energy [28], access to energy and technological benefits [29,30], inability to heat the house [31,32], modern energy technologies [22,30], sources of household energy [33,34] are the most frequently proxies for energy poverty.

Our paper adopts the approach of Papada and Kaliampakos, using population unable to keep warm and arrears on utility bills as measures of energy poverty [35]. We also consider arrears on utility bills for households with dependent children, as these households are expected to be more vulnerable compared to households without dependent children.

Energy poverty measures are correlated with various economic and social indicators. Previous studies explained energy poverty as a lack of enough energy efficiency, low incomes, and energy prices. For example, Halkos and Gkampoura discussed energy poverty in Europe before and during the economic crisis [36]. Churchill et al. connects energy poverty with subjective well-being in Australia [37]. Churchill and Smith tackles energy poverty in the context of ethnicity and mediating role of trust in Australia [32]. Wang et al. consider the role of financial inclusion in the description of energy poverty in China [38]. Siksnelyte-Butkiene et al. connect energy poverty with carbon transition in Greece and Lithuania [39]. Poor people and women are specific categories that require deep analysis in the framework of energy poverty as Awan and Bilgili claimed [40].

In this study, we focus on the impact of income inequality on energy poverty. We selected a limited number of explanatory variables to reflect economic and social characteristics in the analyzed countries.

We consider the Gini index to proxy income inequality, GDP per capita, and renewable energy consumption (REC). We hypothesize that income inequality will have a direct effect on energy poverty indicators.

In addition, REC is expected to reduce energy poverty. Previous studies from literature suggest mixed evidence on the impact of these determinants, as reflected in Table 1.

The EU funds attracted by V4 countries can influence energy poverty in more manners. First, these funds can bring additional capital for renewable energy projects, lowering dependence on expensive traditional fuels. Second, these funds might allow the introduction of advanced technologies that improve energy efficiency, reducing household energy consumption. Third, funds invested in renewables can create jobs, boosting incomes and improving households' ability to afford energy bills.

These EU funds are relevant for V4 countries, because these states do not have enough domestic resources to fully fund the transition to renewable energy. This opportunity can accelerate the transition to renewables compared to relying solely on domestic resources.

The European Union utilizes various financial instruments to address energy poverty across member states. Some key mechanisms and

Table 1
Few determinants of energy poverty in previous studies.

Determinant of energy poverty	Energy poverty indicators	Author(s)	Region	Period	Method	Empirical results
Economic growth/ Economic development	Multidimensional Energy Poverty Index	Acharya and Sadath [41]	India	1999–2008	Correlation method	Negative correlation
	Access to electricity	Amin et al. [42]	States from South Asia (India, Bangladesh, Pakistan, Nepal, Bhutan, Maldives, and Sri Lanka)	1995–2017	Correlation method	Positive correlation
	households that cannot keep homes warm, households with arrears on utility bills, Shares of residential energy expenditure	Bardazzi et al. [43]	Italian regions	2004–2015	beta regression model	Growth has an indirect effect on energy poverty.
	population with access to clean fuels and technologies for cooking purposes, urban/rural/total population having access to electricity, electric power use per capita	Nguyen and Nasir [11]	51 countries	2002–2014	Two sages and three stages least square, two-step system GMM	Negative impact of growth on energy poverty
	population accessing clean fuels and technologies, total/ urban/rural population accessing electricity	Nguyen and Su [30]	56 developing countries	2002–2015	Two-step system GMM and its updated estimate known as SELPDM	Positive impact of economic growth on energy poverty
	Survey data for accessibility to electric appliances, reliability, affordability	Igawa and Managi [10]	37 countries		three-level hierarchical model	Negative impact of growth on energy poverty
Gini index	households who cannot afford to have their homes warm enough	Galvin [44]	EU-28 countries	2009–2017	multivariate statistical analyses.	Asymmetric and direct effect of Gini index on households who cannot afford to have their homes warm enough
	households that cannot keep homes warm, households with arrears on utility bills, Shares of residential energy expenditure	Bardazzi et al. [43]	Italian regions	2004–2015	beta regression model	Income inequality has a direct impact on energy poverty.
	single parent households that cannot heat their homes	Sunikka-Blank and Galvin [45]	EU-28	2010–2018	OLS regressions	Positive impact of energy inequality on energy poverty
	population with access to clean fuels and technologies for cooking purposes, urban/rural/total population having access to electricity, electric power use per capita	Nguyen and Nasir [11]	Italian regions	2002–2014	Two sages and three stages least square, two-step system GMM, Dumitrescu- Hurlin test for causality	Positive effect of Gini index pre-tax and pre-transfer and Gini index post-tax and post-transfer on energy poverty Income inequality causes energy poverty
	population accessing clean fuels and technologies, total/ urban/rural population accessing electricity	Nguyen and Su [30]	56 developing countries	2002–2015	Two-step system GMM estimate	Negative impact of Gini index on energy poverty
	Survey data for accessibility to electric appliances, reliability, affordability	Igawa and Managi [10]	37 countries	2015–2017	three-level hierarchical model	negative impact of low income is heterogeneous across countries.

Source: authors' synthesis.

national plan considerations are presented below:

A. EU Funding Instruments:

- **European Social Climate Fund (SCF):** Established in 2023, this fund specifically allocates resources to EU countries for supporting vulnerable households, including those struggling with energy poverty. The funding supports investments in improving energy efficiency, ultimately reducing energy bills [46].
- **European Structural and Investment Funds (ESIF):** This umbrella program offers funding for various initiatives, including those promoting energy efficiency in housing and renewable energy sources. These improvements can significantly reduce energy consumption and costs for low-income households [47].
- **Horizon 2020 (now Horizon Europe):** The EU's research and innovation program previously funded projects like "Solutions to Tackle Energy Poverty" (STEP) which explored behavioral changes and low-cost solutions to alleviate energy poverty. This type of funding supports the development of innovative approaches to the issue [48].
- **European Energy Efficiency Fund (EEF):** Launched in 2018, this specific fund provides support for energy-saving investments that contribute to reducing energy poverty alongside environmental benefits [49].

B. National Plans:

While the EU provides funding and guidance, national governments have the primary responsibility for tackling energy poverty. The national plans typically address:

- **Timeline:** National plans usually align with the EU's budgetary cycles, often running for several years (e.g., 7 years). They outline a clear timeframe for achieving specific goals.
- **Indicators:** National plans establish measurable indicators to track progress on energy poverty reduction. These indicators include:
  - reduction in the percentage of households experiencing arrears on utility bills (including the subcategory you mentioned - households with dependent children);
  - increase in the energy efficiency of housing stock;
  - number of households receiving energy poverty assistance programs.
- **Specific Measures:** National plans detail specific measures to address energy poverty. These measures refer to:
  - subsidies or grants for energy efficiency renovations in low-income housing;
  - income-based assistance programs to help vulnerable households pay their energy bills;
  - public awareness campaigns to promote energy-saving behaviors.

In most of the studies, more economic growth reduces energy poverty [41,43], but contrary evidence is bought by Amin et al. for some South Asian countries in the period 1995–2017 [42] and by Nguyen and Su for 56 developing countries in the period 2002–2016 [30]. Investment in renewable energy may positively [11] or negatively impact energy poverty [30], while income inequality may enhance [11,43–45] or decrease energy poverty [10,30]. Specific studies consider that vulnerable households have more chances to face energy poverty. For example, income inequality enhances the lack of ability of single parents' household to heat the home in the EU-28 [45].

3. Data and methodological framework

The energy poverty (EP) indicators employed in this paper are provided by Eurostat and refer to:

- Total arrears on utility bills (%).
- Arrears on utility bills households with dependent children (%).
- Population unable to keep home adequately warm (%).

There is a strong correlation between the total arrears on utility bills and arrears for households with dependent children. However, there are still good reasons why we consider them as two separate indicators:

1. **Highlighting Vulnerability:** Looking at the subcategory of households with dependent children specifically draws attention to a potentially more vulnerable group within the population facing energy poverty. These households might have additional financial strains due to childcare or other expenses, making them more susceptible to falling behind on bills.
2. **Policy Targeting:** By separating the data, we suggest that specific policies or assistance programs could be targeted towards this vulnerable group. For instance, there could be initiatives focused on helping families with children keep up with energy bills.
3. **Severity of the Issue:** A high percentage of arrears in the general population is concerning, but a much higher percentage for households with children could indicate a more severe problem. It emphasizes the potential impact on child well-being when families struggle to afford basic utilities.

The indicators used in the models to explain the dependent variable represented by EP are:

- Measure of income inequality: Gini index (the measure to which the repartition of income fails the perfectly equal distribution in a country) from World Bank (index).
- Renewable energy consumption (REC) from World Bank (% of total energy consumption).

The panel data refer to the period 2005–2022 and V4 countries. All the calculations are made in Stata 15. According to Table 2, the maxim economic growth was achieved by Hungary in 2019, while the minimum was reached by Slovakia in 2005. Minimum total arrears on utility bills and arrears on utility bills households with dependent children were observed in Czech Republic in 2019 and maximum values appear in Hungary in 2013. Poland was the state facing maximum population unable to maintain home warm enough in 2019.

Let us consider the general model specification to explain EP (energy poverty):

$$EP_{it} = f(GDP\ per\ capita_{it}, Gini\ index_{it}, REC_{it})$$

To account for panel heterogeneity, the MG estimator takes an OLS approach, involving constructing time-series regressions for each of the 4 countries. The average slope coefficients are then calculated to represent the pooled relationship across countries. This approach acknowledges that coefficients and error variances may vary across

Table 2
Descriptive statistics for V4 countries in the period 2005–2022.

Variable in natural logarithm	Mean	Standard deviation	Minimum	Maximum
GDP per capita	12.05809	2.149166	9.239717	15.25094
Arrears on utility bills, total	2.085942	0.7028897	0.5877867	3.218876
Arrears on utility bills households with dependent children	2.293715	0.6974655	0.8754687	3.493473
Population unable to keep home adequately warm	2.002373	0.5813532	0.7884573	3.514526
GINI index	3.356196	0.1063566	3.206803	3.577948
REC	2.387667	0.2652598	1.841486	2.843903

Source: own calculations.

countries. To ensure the validity of the MG estimator, preliminary tests for cross-sectional dependence, heterogeneity, and unit roots are conducted. MG is the best choice in this study due to flexibility (MG allows for heterogeneity across groups (individuals, firms, etc.) in the panel), MG helps control for these unobserved effects by allowing for group-specific intercepts). However, if the dataset is not large the results based on MG estimator might be less reliable.

Cross-sectional dependence reflects a situation where the error terms (unexplained portions) of different observations in a panel data model are correlated with each other. This means the outcome variable for one country is not only influenced by its own independent variables but also by the unobserved factors affecting other countries in the data set.

Cross-sectional dependence tests are crucial tools in panel data analysis because they address a potential issue that can significantly affect the reliability and interpretability of results. For example, spatial dependence and common effects might generate cross-sectional dependence.

- **Unobserved common shocks:** Events like recessions, financial crises, or natural disasters can simultaneously affect multiple individuals in a panel.
- **Spatial dependence:** Individuals residing in close proximity (e.g., neighboring countries or states) might be influenced by each other's economic activities.

If these dependencies are not accounted for, the standard errors of the estimated coefficients will be underestimated. This leads to inflated confidence intervals and potentially misleading conclusions about the significance and magnitude of the estimated relationships. Ignoring cross-sectional dependence generates overestimating the effectiveness of policies in one region if they indirectly affect others.

Panel data may exhibit heterogeneity, meaning that countries within the panel may behave differently. Some tests account for this heterogeneity, while others assume a common behavior across units, and this choice can affect the results.

Unit root tests of the second generation like CADF should be applied before estimations. A series with a unit root, meaning it does not mean-revert over time, can lead to "spurious regressions". This occurs when two non-stationary series appear to have a statistically significant relationship, even though there is none in reality. Unit root tests help identify such non-stationary series, allowing you to avoid drawing false conclusions from your analysis. In some models the use of non-stationary data series with only one unit root might be allowed, but these tests should be applied to ensure that no more unit roots are present.

4. Results and discussion

Cross-sectional dependence is checked for all the data series according to Pesaran CD test at 10 % significance level and is expected due to similar evolutions of the indicators for countries in the same region (see Table 3).

The CADF test is applied on panel data to check for unit root. Knowing that this particular test generates results that depend on the number of lags, two versions are considered. The equation for data series in level includes intercept and trend, but only intercept is kept in the equation for data in the first difference (see Table 4).

Table 3
The results of cross-sectional dependence tests.

Variable in natural logarithm	Pesaran CD
GDP per capita	8.95 (<0.01)
Total arrears on utility bills, total	4.53 (<0.01)
Arrears on utility bills households with dependent children	4.35 (<0.01)
Population unable to keep home adequately warm	11.30 (<0.01)
GINI index	1.67 (0.094)
REC	8.68 (<0.01)

Source: authors' calculations; *p*-values are displayed in brackets.

The results of CADF test show that all the data series are stationary in the first difference at 5 % significance level, but non-stationary in level. Therefore, the data series are integrated of order 1. The matrix of correlation suggested non-significant correlations between explanatory variables in the models.

According to Table 5, GDP per capita reduced energy poverty in V4 countries in the period 2005–2022. Renewable energy consumption enhanced arrears, but reduced the population unable to keep warm. Income inequality measured by Gini index and FDI enhanced energy poverty. By adding one more explanatory variable in the models (Gini index), the adjusted R-squared increased, which supports the hypothesis that Gini index has significant contribution in explaining energy poverty.

A variety of explanatory factors exhibit differing impacts on various energy poverty indicators, including total utility bill arrears, arrears in households with dependent children, and the proportion of the population unable to maintain adequate home warmth. GDP per capita has an indirect impact on all EP indicators. This suggests that improved economic conditions and higher income levels enhance individuals' financial standing, enabling them to afford energy for their fundamental needs, such as paying utility bills and keeping their homes warm. Our findings align with those of Rodríguez-Alvarez et al. and Zhao et al., who also confirmed the negative impact of per-capita income on energy poverty [12,50].

Regarding the Gini index, the findings suggest that income inequality elevates all EP indicators. Income inequality is considered one of the primary drivers of growing energy poverty, as those with higher incomes can afford energy and install solar power systems, while low-income households are frequently denied these advantages. Low-income households have maximum chances to encounter energy poverty; however, the severity varies and depends on the national income level [10]. The majority of researchers in the existing literature unanimously agree that income inequality exacerbates energy poverty. For instance, McGee and Greiner and Nguyen and Nasir affirmed that more income disparity intensifies energy poverty [11,51].

Renewable energy use has a heterogeneous impact on various EP indicators. The results suggest that renewable energy has a negative effect, reducing the percentage of the population unable to have warm in buildings. The results further indicate that renewable energy consumption promotes total utility bill arrears and arrears in households with dependent children. This may be attributed to the higher cost of renewable energy across the EU, largely due to policies such as FIT (feed-in tariffs) and RPS (renewable portfolio standards) [52]. In these instances, governments provide financial subsidies as well as above-market tariff rates to assist companies in developing new renewable energy projects and minimizing their investment risk. We should also take into account that costs of proper subsidies and incentives are often passed on to end consumers in the form of taxes and surcharges, pushing consumers further behind.

Income inequality is a major factor contributing to energy poverty. Therefore, government should provide financial aid directly to low-income households to help them afford energy bills, offer tax rebates or bill reductions for low-income households and fund programs that improve energy efficiency in poor households, reducing energy consumption and lowering bills. V4 should implement a system where low-income households pay a lower rate for a base level of energy consumption, with prices increasing for higher usage. Moreover, they should regulate energy company profits to ensure they're not excessive, keeping energy costs fairer.

5. Conclusions

The pervasive issue of energy poverty (EP) continues to pose significant challenges for the V4 countries, affecting a substantial portion of

Table 4
The results of CADF test.

Variable in natural logarithm	Data in level		Data in the first difference	
	One lag	Two lags	One lag	Two lags
GDP per capita	0.356 (0.639)	−0.472 (0.318)	−3.024 (<0.01)	2.968 (<0.01)
Total arrears on utility bills	0.093 (0.537)	−0.949 (0.171)	4.226 (<0.01)	3.753 (<0.01)
Arrears on utility bills households with dependent children	−0.560 (0.288)	0.853 (0.803)	−2.964 (<0.01)	−3.046 (<0.01)
Population unable to keep home adequately warm	1.508 (0.934)	2.579 (0.995)	3.204 (<0.01)	4.385 (<0.01)
Gini index	2.945 (0.998)	−1.130 (0.129)	−3.745 (<0.01)	2.996 (<0.01)
REC	−1.627* (0.052)	1.011 (0.844)	−2.677 (0.004)	−3.567 (<0.01)

Source: authors' calculations; *p*-values are displayed in brackets.

Table 5
Models to explain energy poverty based on renewable energy consumption in V4 countries.

Variable	Dependent variables											
	M1 Total arrears		M2 Total arrears		M3 Total arrears		M4 Arrears for households with dependent children		M5 Population unable to keep home warm		M6 Population unable to keep home warm	
	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
GDP per capita	−3.746249	<0.01	−3.84666	<0.01	–	–	−3.623641	<0.01	−2.988005	<0.01	−2.137923	0.014
REC	0.9231311	0.001	0.8777467	0.001	0.1982937	<0.01	0.8843818	0.005	−1.1748193	0.0215	−0.3613898	0.092
Gini index	–	–	2.452594	0.0611	5.922678	0.051	–	–	–	–	2.605447	<0.01
Constant	45.02698	<0.01	48.28501	0.001	–	<0.01	43.99334	<0.01	39.93894	<0.01	22.2668	0.030
Adjusted R-squared	0.772		0.779		0.773		0.704		0.712		0.715	

Source: authors' calculations.

households across the member states. This study showed a complicated connection between EP and income inequality, employing a comprehensive analysis of three distinct EP indicators across the V4 countries from 2005 to 2022.

The findings reveal a multifaceted relationship between EP and income inequality, with both direct and indirect impacts. Economic growth per capita emerges as a key determinant of EP reduction, as higher GDP levels enhance household incomes, enabling greater access to affordable energy resources.

The Gini index, reflecting income inequality, demonstrates a positive and significant association with all EP indicators. This highlights the disproportionate burden of energy costs in poor homes, exacerbating their struggles to meet basic energy needs.

Renewable energy consumption, while generally recognized for its environmental benefits, presents a paradoxical impact on EP. While renewable energy reduces population unable to keep home warmly, it simultaneously contributes to rising arrears on utility bills. This paradoxical effect can be attributed to the associated high taxes and surcharges associated with renewable energy deployment.

The findings of this study provide valuable insights for policymakers seeking to mitigate energy poverty and promote sustainable energy transitions:

- Address income inequality: Implementing policies that promote equitable access to employment opportunities can effectively reduce income inequality and consequently alleviate energy poverty.
- Promote more investment in renewable energy projects: Attracting investment in renewable energy projects can accelerate the deployment of sustainable energy sources while generating economic benefits that can contribute to alleviating energy poverty.
- Reformulate renewable energy policies: Designing renewable energy policies that minimize the burden of renewable taxes and surcharges on end-users can shield vulnerable households from the potential price spikes associated with renewable energy transition.

All of these policies should help promote a more sustainable environment and help lead to achieving the Sustainable Development Goals. While this study provides valuable policy recommendations, it acknowledges limitations that can serve as research topics for future studies. Specifically, future research should:

- Explore the bidirectional relationship between EP and income inequality: Conducting causality analysis can provide a deeper understanding of the causal pathways between these two factors.
- Analyze the differences between V4 countries and other EU states: A comparative analysis can identify specific factors that contribute to EP in different member states and tailor policy interventions accordingly.
- Investigate the effectiveness of targeted EP interventions: Assessing the impact of specific EP mitigation programs can inform evidence-based policy formulation.
- Address energy efficiency: Examining the impact of energy efficiency solutions on reducing energy consumption and alleviating EP can complement broader policy efforts.
- Consider the role of institutional factors: Assessing the influence of institutional structures, such as regulatory frameworks and energy market design, can identify areas for improvement.

By considering these gaps, future research will certainly deepen our understanding of the complex relationship between EP and income inequality, leading to more effective and targeted policy interventions to alleviate EP and promote sustainable energy transitions in the V4 states. The study is limited to Gini index as proxy for income inequality. In a future study, more indicators will be considered (gender pay gap, Theil index etc.).

CRediT authorship contribution statement

Mihaela Simionescu: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Javier Cifuentes-Faura:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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